ERRATUM TO VERSION 1 OF "MORSE-BASED MODULAR HOMOLOGY FOR EVOLVING SIMPLICIAL COMPLEXES" (ARXIV:2508.14429V1)

A PREPRINT

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ABSTRACT

Version 1 of arXiv:2508.14429 maintained Betti numbers under local edits by re-reducing, after each edit, only the critical columns of a discrete-Morse reduced boundary matrix that the edit affects directly. It stated that this update is exact. That claim is withdrawn: the rule is not exact, for two independent reasons. It re-reduces a replaced column but not the reduced columns that had absorbed it (E1), and its set of affected critical columns misses columns whose Morse boundary changes through gradient paths that end far from the edit (E2). This note states what version 1 specified, gives instances on which its rule fails although none of its safeguards applies, records an inconsistency in its experiment P2, and measures how often an ablation that isolates E1 is wrong. The current version of arXiv:2508.14429 uses a different algorithm, the dynamic reduced basis, whose exactness is proved there.

1 The update rule of version 1

Section and figure numbers prefixed by "§" refer to version 1 [1]; references to theorems, definitions and sections of the current version are marked as such.

What version 1 specified. An acyclic matching on K defines critical cells, critical chain groups C_k^C and a critical boundary ∂_k^C "induced by the matching via V-paths (Forman's gradient flow)", with matrices B_k^C. Betti numbers are read off from $r_k = \operatorname{rank} B_k^C$ and the numbers n_k^C of critical cells (§2.2, §2.5). The matching is computed by a discrete Morse reduction at initialisation and again at *recompression events* (§2.2), which are triggered by any of: the periodic trigger $t \equiv 0 \pmod m$; the locality trigger $\rho_t \geq \tau$, where

$$\rho_t = \frac{\sum_k |R_k^C(t)|}{\max\bigl(1, \sum_k |C_k^C(t)|\bigr)},$$

$C_k^C(t)$ being the critical k-cells before update t and $R_k^C(t) \subseteq C_k^C(t)$ those "whose column incidences change due to update t"; and a validity trigger when a precondition of a combinatorial gate fails. Typical values are $m = 32$ and $\tau = 0.30$. The gates are optional ("No combinatorial gate is used in P2", §3.3). No rule is given for updating the matching between recompressions. For an edit, $R_k \subseteq K_k$ is the set of k-simplices "whose incidences in the boundary operators are affected by the modification", computed "via a face-difference procedure that compares the pre-update

and post-update stars of the modified region", and the affected critical columns are

$$R_k^C = \{\, j : \sigma_j^k \in C_k^C \text{ and } \sigma_j^k \in R_k \,\}$$

(§2.3). The columns $\{B_k^C[:, j] : j \in R_k^C\}$ are replaced by their new incidence patterns, "followed by reapplying column-reduction operations only to these affected columns and to any columns whose pivots conflict with them"; "when a column is replaced, its pivot assignment is cleared, the column is re-reduced against existing pivots, and any resulting pivot conflicts are propagated in a localized manner". Two further clauses concern rows and ranks. §2.3: "If the update inserts or deletes $(k-1)$-simplices (thus changing the row set of B_k^C), the pivot map of B_k^C is re-initialized for that dimension." §2.5: r_k is recomputed "if either the column set of B_k^C changes ($R_k^C \neq \emptyset$) or the row set of B_k^C changes", and "if a localized reducer is available in dimension k^*, the rank r_{k^*} is updated incrementally by reassigning pivots in the column-reduced form (over $\mathbb{Z}_2$), avoiding any operation on unaffected columns". Version 1 states that its rule "ensures that the rank of B_k^C is updated exactly" (§2.3) and "guarantees exact Betti numbers" (§2.5). Both statements are false, for two independent reasons, E1 and E2.

What counts as a counterexample. An instance refutes version 1 as specified only if none of its safeguards applies. All instances below have at most four updates, so the periodic trigger does not fire. For the locality trigger we bound ρ_t from above with the most generous numerator that any local reading of $R_k^C(t)$ allows: every cell that is critical before *or* after update t and lies in the closed star of a vertex of an edited simplex, before or after the update; the denominator is the number of critical cells before the update, as defined above. We require this bound to be below $\tau = 0.30$ at every update. The gate preconditions (connected 2-manifolds) fail on these complexes, so with gates enabled the validity trigger would request a recompression; the instances therefore use version 1 without gates, which is how it ran P2. Finally, we check each instance against the row and rank clauses quoted above.

E1: a reduced column records the columns it absorbed. In a column reduction $R = BV$, a reduced column $R_b = \partial b + \partial a_1 + \dots$ is correct only while the absorbed columns a_i are present with their old boundaries. The rule above re-reduces a replaced column a, but leaves every R_b that absorbed a in place. Such an R_b need not own a pivot (it may be zero), so it is never in conflict with anything, and nothing in the rule revisits it. A deletion is a replacement by the zero column and fails in the same way. The dynamic reduced basis (DRB) of the current version repairs exactly this: $\mathrm{occ}(a)$ lists every slot whose V contains a, and DELETE rewrites all of them (Theorem 1(c) of the current version).

Example 1.1 (E1, matrix form)**.** Rows $e_1 < e_2$, $B = [\, a \mid b \,]$ with $a = b = e_1$. Reduction: $R_a = e_1$ owns pivot e_1; $R_b = b + a = 0$; rank 1. Replace a by e_2: its pivot is cleared, it is re-reduced to $R_a = e_2$, which owns the free pivot e_2, so there is no conflict; $R_b = 0$ is not touched. The reported rank is 1; the rank of $[\, e_2 \mid e_1 \,]$ is 2.

Example 1.2 (E1 for version 1 as specified)**.** Let K_0 consist of 20 isolated vertices $0, 1, \dots$. Every matching on K_0 is empty, since K_0 has no pair of incident cells, so the Morse reduction at initialisation leaves every cell critical. Version 1 gives no rule for matching inserted cells; we keep them critical, so $B_k^C = B_k$ throughout and the localized reducer acts on B_1 itself. Apply four updates of one edit each, with rows and columns in index order:

$$t = 1\colon +\{0,1\}, \qquad t = 2\colon +\{0,2\}, \qquad t = 3\colon +\{1,2\}, \qquad t = 4\colon -\{0,1\}.$$

After $t = 3$ the reduction gives $R_{01} = \{0,1\}$, $R_{02} = \{0,2\}$ and $R_{12} = \{1,2\} + \{0,2\} + \{0,1\} = 0$. At $t = 4$ the column of $\{0,1\}$ releases its pivot and is dropped, and $R_{12} = 0$ stays. The reported Betti numbers are $(\beta_0, \beta_1) = (19, 1)$; the truth is $(18, 0)$, because $\{0,2\}$ and $\{1,2\}$ form a path. The DRB has $V_{12} = \{\{1,2\}, \{0,2\}, \{0,1\}\}$, so $\mathrm{occ}(\{0,1\})$ contains the slots of $\{0,1\}$ and $\{1,2\}$; Case A sets $V_{01} \leftarrow \{\{1,2\}, \{0,2\}\}$ (still $\partial V_{01} = \{0,1\} = R_{01}$), removes the slot of $\{1,2\}$, and reports $(18, 0)$. *Safeguards.* The bound on ρ_t is 0.15, 0.24, 0.27 and 0.26 for $t = 1, \dots, 4$ (at most 0.27). No update inserts or deletes a vertex, and no vertex changes its critical status, so neither the literal row-set clause nor a broader reading of it applies to B_1^C. The failure does not depend on the order of the rows $0, 1, 2$: it occurs for 6 of the 6 orders. *The rank clause.* If "r_k is recomputed" in §2.5 is read as a recomputation of $\mathrm{rank}\, B_1^C$ from scratch whenever a column of B_1^C changes, then E1 does not occur, but every such update is then a full rank computation of B_1^C. Version 1 presents the localized reducer, which updates r_k "incrementally by reassigning pivots", as its method; E1 refutes that reducer. E2 below fails under every reading.

Without the padding, the same seven edits ($+\{0\}, +\{1\}, +\{2\}$ followed by the four updates above) are the shortest edit sequence on which the rule fails. An exhaustive enumeration of the valid edit sequences on at most four vertices, up to relabelling (1, 2, 5, 15, 61 and 263 sequences of lengths 1 to 6, and 1,250 of length 7), finds no failure of length at most 6 and 18 of length 7; a sequence of length at most 6 on five or more vertices contains at most one edit of positive dimension and cannot fail. On so small a complex, however, ρ_t exceeds τ, which is why the instance above is padded.

E2: the affected set misses distant critical columns. The critical boundary is a sum over gradient paths, $\partial^M c = \Pi(\partial c)$ (Section 4 of the current version), and it depends on every pair along every gradient path leaving the facets of c. These paths can be arbitrarily long, so an edit to a matched cell can change $\partial^M c$ for critical cells c far from the edit, while R_k^C contains only critical cells whose own incidences change. Since version 1 does not say how the matching is maintained between recompressions, we use the least intrusive convention: keep every pair that does not involve a deleted cell; the partner of a deleted matched cell, and every inserted cell, becomes critical. The restriction of an acyclic matching is acyclic, so this is a valid matching. (Under this convention insertions are harmless: a new cell is maximal and critical and lies on no gradient path. Deleting a matched cell is not.) In the two examples below the stored column of the distant critical cell is never replaced, so the error is the same under the three readings of the row and rank clauses: re-initialising the pivot map rebuilds it from the stored, stale columns; recomputing r_k computes the rank of the stored, stale matrix; and the incremental reducer only inserts the column of the new critical cell.

Example 1.3 (E2 on a graph)**.** Let K_0 be two disjoint 8-cycles, on the vertices $0, \dots, 7$ and $8, \dots, 15$, with the collapse matching of Definition 4.1 of the current version. Its critical cells are the edges $c = \{0, 1\}$ and $\{8, 9\}$ and the vertices $r = \{5\}$ and $\{13\}$, and $\partial^M c = r + r = 0$: the gradient paths from the two endpoints of c are the two arcs of the first cycle from those endpoints to r. At $t = 1$ delete the matched edge $y = \{2, 3\}$, whose partner is $\{2\}$. Now $\{2\}$ is critical, the arc through y ends at $\{2\}$, and $\partial^M c = \{2\} + \{5\}$. The edge c shares no vertex with y, so its column is not in R_1^C and stays 0. The reported Betti numbers are $(3, 2)$; the truth is $(2, 1)$. *Safeguards.* The closed stars of the vertices 2 and 3 contain no critical cell before the update, and after it only $\{2\}$, so $\rho_1 \le 1/4 = 0.25$. No vertex is inserted or deleted, so the literal row-set clause does not apply to B_1^C; under a broader reading the new critical row $\{2\}$ re-initialises the pivot map of B_1^C, which leaves the stored column of c at 0. More generally, on a cycle C_L the collapse leaves one critical edge c and one critical vertex r, the two arcs from the endpoints of c to r contain every matched edge exactly once, and deleting any matched edge $y = \mu(x)$ makes $\partial^M c = r + x \neq 0$; when y avoids c and r, version 1 does not see the change. A second component (or padding by isolated vertices) is needed only to keep ρ_t below τ.

Example 1.4 (E2 with a critical triangle)**.** Let K_0 be the 5×5 grid torus on the vertices $1, \dots, 25$, in which vertex (i, j), $i, j \in \mathbb{Z}/5$, is labelled $1 + 5i + j$ and the triangles are

$$\{(i,j),(i{+}1,j),(i{+}1,j{+}1)\}, \qquad \{(i,j),(i,j{+}1),(i{+}1,j{+}1)\},$$

together with the cone with apex 0 over the meridian 1–6–11–16–21; $\beta(K_0) = (1, 1, 1)$. The collapse matching (first-in-first-out queue) has the critical cells $\{0, 1, 6\}$, $\{1, 2, 7\}$, $\{4, 10\}$, $\{15, 16\}$ and $\{24\}$, and $\partial^M\{1, 2, 7\} = 0$. At $t = 1$ delete the matched triangle $\{8, 13, 14\}$, paired with the edge $\{8, 13\}$. This changes $\partial^M\{1, 2, 7\}$ to $\{8, 13\}$, although $\{1, 2, 7\}$ and $\{8, 13, 14\}$ share no vertex. The reported Betti numbers are $(1, 2, 1)$; the truth is $(1, 1, 0)$. *Safeguards.* The only cell that is critical before or after the update and lies in the closed stars of 8, 13 and 14 is $\{8, 13\}$, so $\rho_1 \le 1/5 = 0.20$. The update deletes a 2-simplex, so the literal row-set clause concerns B_3^C, which has no columns; under a broader reading the new critical row $\{8, 13\}$ re-initialises the pivot map of B_2^C, which leaves the stored column of $\{1, 2, 7\}$ at 0.

Smaller complexes show the same change of $\partial^M c$, for instance a single 4-cycle or a 3×3 torus with a disk, but on them $\rho_1 \ge \tau$ under the counting above, so version 1 would recompress and return correct Betti numbers. They illustrate the defect of the affected set but do not refute version 1 as specified.

Measured error rates. Section 2 measures defect E1 in isolation, on the workloads of the current version, with and without version 1's periodic recompression.

P2: an inconsistent workload description. Version 1, §3.3, defines the P2 sequence as alternating a refinement and "its exact inverse", REFINE(R_t) followed by COARSEN(R_t). Under that definition $|K_t|$ is bounded and returns to $|K_0|$ after every coarsening. The same section states that each step injects $O(1)$ new simplices, "thus $|C_t| = \Theta(t)$", and Figures 4 and 5 of version 1 show sizes growing linearly over 500 updates. The two descriptions cannot hold for the same sequence, and version 1 does not determine which sequence produced its P2 timings. The workload W2 of the current version lets the size perform a random walk around its initial value and records it.

What the experiments of version 1 checked. Version 1 shows Betti traces for P1 and P3 (its Figures 2(b) and 7(b)) and states that they match the specification (P1) and agree across methods (P3). Both are connected surfaces with known connectivity, whose Betti numbers follow from χ, $\beta_0 = 1$ and the boundary-edge criterion without any matrix (Remark 5.1 of the current version), the formulas behind version 1's gates; version 1 does not state whether its gates were enabled there. For P2, the one experiment run without gates, version 1 reports no Betti numbers and does not compare its output with an independent computation. Such a comparison would have exposed E1 even though the Betti numbers of P2 are constant: on the analogous workload W2, lcr reports wrong Betti numbers after almost every update, for the first time at update 1 (Section 2, Table 1). In the current version every exact method is compared with independent recomputations (Section 6.1 of the current version).

Cost claims. Version 1, §2.6, states a local reduction cost of $\Theta\big(\sum_k (s_k^C)^2\big)$ "under sparse operations with limited fill-in". That analysis concerns the rule shown above to be inexact, and we make no statement about it. For the exact method of the current version, Proposition 4 of the current version shows that no bound in terms of the size of the update holds, and we report costs as measurements.

2 Measured error rates of the rule without basis tracking

Ablations. lcr applies version 1's re-reduction rule to the full boundary matrices without basis tracking: an inserted column is reduced against the existing pivots, and a deleted column releases its pivot and is dropped. It runs in the engine of the current version, uses exactly the order of its method mmhm-norb (the Morse order of K_0, no rebuilds) and differs from it only in not storing V and the footprints, so its errors are caused by defect E1 alone. It is not a reimplementation of version 1: it forms no Morse complex (so E2 cannot occur), never recompresses and has no gates. lcr-rc adds version 1's periodic recompression: every $m = 32$ updates, version 1's default, it recomputes all reduced columns from scratch in the current key order by the static reduction with clearing. It models only this trigger, without a Morse complex, the ρ trigger or gates. Both ran untimed on one instance of each workload variant of the current version (Section 6.1 there), three runs each, and were compared with a static recomputation after every update; the *error rate* is the fraction of updates after which at least one Betti number is wrong. These rates measure E1, not version 1 as a whole: on the connected surfaces of W1, gates of the kind version 1 describes would return correct Betti numbers (Remark 5.1 of the current version).

Results. lcr is wrong after 99.3 % to 100.0 % of the updates of every W1–W4 instance; on W2 after 5,999 of 6,000. Its first error comes within the first 16 updates of every run in which it errs, and on W1–W4 it stays wrong after almost every later update, so these percentages mainly say how early the first error comes. The Betti numbers of W2 are constant; the errors there come from the algebra alone, and a comparison with the known $(1, 0, 0, 0)$ detects them. On W5b, lcr is wrong after 50.0 % of the updates: after every deletion of the toggled triangle and after no reinsertion. The zero column that represents the 2-cycle of the sphere absorbed the column of the toggled triangle when K_0 was reduced; lcr never revisits it, so after each deletion it still reports the destroyed 2-cycle, which exists again after each reinsertion. On W5a, lcr is never wrong (0.0 %), and W5a cannot expose defect E1: the cycle graph has no free face, so the collapse first removes the edge with the smallest vertex tuple, $\{0, 1\}$, as a critical cell, which gives it the largest key; W5a toggles exactly this edge, and every reinsertion gives it the largest key again. No other column ever absorbs the column with the largest key, so dropping it leaves a correct reduction. Pooled over these instances, lcr is wrong after 83.1 % of the updates; this number depends on the mix of instances.

lcr-rc coincides with lcr until its first recompression, and its first errors come at the same updates (Table 1). A recompression is a static reduction, so it makes the Betti numbers correct again. Pooled over the same instances, lcr-rc is wrong after 60.2 % of the updates: after 2.8 % on W2 (168 of 6,000), 0.8 % on W5b and 0.0 % on W5a, but after 84.5 % to 93.8 % on W1, W3 and W4, where the errors return soon after each recompression. On W2, the analogue of version 1's P2, recompression thus hides most errors, but a comparison with the known Betti numbers still finds them from the first updates on.

References

[1] Anqiao Ouyang. Morse-based modular homology for evolving simplicial complexes. arXiv preprint arXiv:2508.14429v1 [cs.CG], 2025. Version 1.

Table 1: Upper part: correctness of the exact methods of the current version (as in its correctness table). Lower part: the ablations lcr and lcr-rc on one instance of each workload variant, three runs each: updates, the percentage of updates after which each was wrong, and the first wrong update (minimum over the runs).

method	runs	states	vs. full-id	vs. analytic	vs. GUDHI	mismatches
full-id	126	252,126	(ref.)	210,105	203,526	0
full-sorted	69	138,069	138,069	96,048	111,069	0
full-morse	87	174,087	174,087	132,066	147,087	0
gudhi	126	203,526	203,526	161,505	(ref.)	0
drb-id	126	252,126	252,126	210,105	203,526	0
drb-id-rb	288	576,288	576,288	426,213	527,688	0
drb-rand	12	24,012	24,012	18,009	24,012	0
mmhm-norb	126	252,126	252,126	210,105	203,526	0
mmhm	288	576,288	576,288	426,213	527,688	0
euler	42	84,042	84,042	84,042	57,042	0
all exact	1,290	2,532,690	2,280,564	1,974,411	2,005,164	0

			lcr		lcr-rc ($m = 32$)	
instance	runs	updates	% wrong	first wrong	% wrong	first wrong
W1 sphere $L = 5$	3	6,000	99.5	1	91.0	1
W1 torus 128×128	3	6,000	99.3	1	90.8	1
W2 $N = 15$, $s = 8$	3	6,000	100.0	1	2.8	1
W3 $N = 14$	3	6,000	100.0	1	93.8	1
W4 2D $n = 4$k, $D = 2$	3	6,000	99.6	5	84.5	5
W4 2D $n = 4$k, $D = 3$	3	6,000	99.6	5	87.2	5
W4 3D $n = 4$k, $D = 3$	3	6,000	99.9	2	91.3	2
W5a $L = 10{,}000$	3	6,000	0.0	–	0.0	–
W5b level 5	3	6,000	50.0	1	0.8	1